



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

B.TECH, FOURTH SEMESTER

End Semester Examination - April / May -2025

Mathematical Foundations for Data Science - II [MAT-2213]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Consider a Markov chain with state space $S = \{0, 1\}$, where 0 represents rain and 1 represents no rain; X_n – weather on day n . We assume that:
- A)
- If it rains today, then it will rain tomorrow with probability 0.75,
 - If it does not rain today, then it will rain tomorrow with probability 0.35
- (3)
- What is the probability that it will rain four days from today given that it is raining today?
- B)
- Consider the Markov chain with transition probability matrix
- $$P = \begin{bmatrix} 1/3 & 2/3 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix}$$
- (3)
- Classify the states into recurrent and transient and prove it mathematically.
- C)
- i) The factory has 2 machines and one repair crew. Assume that probability of any one machine breaking down on a given day is α . Assume that if the repair crew is working on a machine, the probability that will complete the repair in a day is β . For simplicity, ignore the probability of repair completion or breakdown taking place except at the end of the day. Let X_n be the number of machines in operation at the end of the n^{th} day. Assume the behavior of X_n can be modeled as a Markov chain.
- (2)
- Write Transition probability matrix for above defined problem.
- ii
- (2)

For sufficiently small $h > 0$, prove the following property of the Poisson process $\{X(t): t \geq 0\}$ with state space $S = \{0, 1, 2, \dots\}$,

$$p_{i,j}(h) = P(X(t+h) = j \mid X(t) = i) = o(h), j \geq i + 2$$

where $o(h)$ is the order of 'h' and satisfies $\lim_{h \rightarrow 0} \frac{o(h)}{h} = 0$, λ is the rate of occurrence of events.

- 2) Prove that the steady state solution of M/M/1 queueing model yields $p_n = \left(\frac{\lambda}{\mu}\right)^n p_0$, where p_n is the probability of having 'n' customers, p_0 denotes the probability of having '0' customers in the system, λ is arrival rate of the customers and μ is service rate of the servers in the system. (3)
- A)
- B) A Commercial bank has 3 cash paying assistants customers are found to arrive in a Poisson fashion at an average rate of 6/hr for business transaction. The service time is found to have an E.D with a mean of 18 mins. The customers are processed on FCFS basis. Calculate average number of customers in the system. (3)
- C) Let $X = \{X_1, X_2\}$ and $Y = \{Y_1, Y_2\}$ be two groups of variables and $Z = (X, Y)^T$. The Covariance matrix of Z is given below.

i

$$\begin{bmatrix} 4.0 & 2.1 & 0.3 & 1.2 \\ 2.1 & 3.5 & 0.4 & 0.9 \\ 0.3 & 0.4 & 2.0 & 0.5 \\ 1.2 & 0.9 & 0.5 & 3.1 \end{bmatrix} \quad (2)$$

- a. Find the Covariance of (X_1, Y_2) .
- b. Find the Variance of Y_1 .

- ii Provided the data of 10 number of observations with its mean vector as

$$\bar{X} = \begin{pmatrix} 4097.5 \\ 1468.2 \\ 3700.1 \end{pmatrix} \text{ and variance-covariance matrix as } S = \begin{pmatrix} 12497245 & 3634519 & 11454489 \\ 3634519 & 1265551 & 3317404 \\ 11454489 & 3317404 & 10817991 \end{pmatrix}, \text{ the } T^2 \text{ statistic calculated as } 26.56274. \quad (2)$$

Test the hypothesis

$$H_0: \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} = \begin{bmatrix} 2000 \\ 1200 \\ 2500 \end{bmatrix} \text{ vs } H_1: \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} \neq \begin{bmatrix} 2000 \\ 1200 \\ 2500 \end{bmatrix}$$

- 3) (4)

Calculate SST, SSR and SSE for the following data

A)

X_1	X_2	Y
15	30	4
12	24	6
8	20	7
6	14	9
4	10	13
3	4	15

With the given regression equation as $\hat{Y} = 16.48 + 0.39 X_1 - 0.62 X_2$.

B)

Find the population covariance matrix for the following table.

Score	Age
68	29
60	26
58	30
40	35

(4)

C)

Consider a one-way MANOVA study with 4 groups and 2 dependent variables. The sum of squares and cross-products matrices are:

- Between-group matrix (B): $\begin{pmatrix} 36 & 12 \\ 12 & 48 \end{pmatrix}$
- Within-group matrix (W): $\begin{pmatrix} 90 & 30 \\ 30 & 100 \end{pmatrix}$

(2)

Calculate Wilks' Lambda statistic.

4)

The following observations on two responses are collected for three treatments. The observation vectors $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ are

A)

Treatment 1: $\begin{bmatrix} 4 \\ 6 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \end{bmatrix}, \begin{bmatrix} 7 \\ 5 \end{bmatrix}, \begin{bmatrix} 5 \\ 8 \end{bmatrix}, \begin{bmatrix} 4 \\ 8 \end{bmatrix}$,

Treatment 2: $\begin{bmatrix} 4 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ 7 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix}$,

Treatment 3: $\begin{bmatrix} 3 \\ 6 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.

(4)

Construct the one-way MANOVA Table.

B)

(4)

The following data set is obtained from a study conducted to determine the effect of job stress on work safety perception. Two measurements ($p=2$), job stress and safety perception were taken on two groups of people with sample size of 19. The mean vectors and covariance matrices are as follows:

$$\text{For group-1: } \bar{X} = \begin{bmatrix} 140 \\ 420 \end{bmatrix}; S = \begin{pmatrix} 550 & 210 \\ 210 & 330 \end{pmatrix},$$

$$\text{For group-2: } \bar{X} = \begin{bmatrix} 150 \\ 400 \end{bmatrix}; S = \begin{pmatrix} 480 & 250 \\ 250 & 530 \end{pmatrix},$$

The management claims that there is no significant difference in job stress and safety perception between the two groups of employees. Does the data substantiate the claim at $\alpha = 0.01$?

- C) Consider a set of 5 one-dimensional data points 50, 51, 55, 57, 60. Apply Agglomerative clustering algorithm to build the hierarchical clustering dendrogram. Merge the clusters using MIN distance. (2)

- 5) Compute the linear discriminant projection of the following two-dimensional dataset:

- A) (4)
- Class C_1 : (4, 1), (2, 4), (2, 3), (3, 6)
 - Class C_2 : (9, 10), (6, 8), (9, 5), (8, 7)

- B) Given the data in the table, reduce the dimension from 2 to 1 using the principal component analysis (PCA) algorithm.

X_1	2	0	3	1
X_2	0	1	1	2

(3)

- C) Consider the data points 30, 35, 32, 44, 31, 29, 17, 26, 48. Use k-means algorithm to cluster the data into 2 groups. The initial cluster centroids are $M_1 = 10$, $M_2 = 20$. The distance function used is Euclidean distance. (3)

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